Tutorial 12

Ques 1

(a) $(\mathbb{Z}, +)$

→ **Closure** :- let $a, b \in \mathbb{Z}$
then $a+b \in \mathbb{Z}$
Hence closed under addition

→ **Associativity** :- let $a, b, c \in \mathbb{Z}$
$(a+b)+c = a+(b+c)$
Hence, associative

→ **Identity** :- There exists $0 \in \mathbb{Z}$ such that,
$a+0 = 0+a = a \quad , \forall \mathbb{Z}$
So, Identity exists.

→ **Inverse** :- For each $a \in \mathbb{Z}$, $\exists -a \in \mathbb{Z}$ such that
$a+(-a) = (-a)+a = 0$
So, Inverse exists

Hence $(\mathbb{Z}, +)$ is a group.

(b) $(\mathbb{N}, +)$ let $a, b, c \in \mathbb{N}$

→ **Closure** :- $a+b \in \mathbb{N}$
Hence, closed under addition

→ **Associativity** :-
$(a+b)+c = a+(b+c)$
Hence, associative

→ **Identity** :- There does not exist an element $e \in \mathbb{N}$ such that
$a+e = a, \forall a \in \mathbb{N} \quad (\because 0 \notin \mathbb{N})$

→ **Inverse** :- For $a \in \mathbb{N}$, there does not exist $b \in \mathbb{N}$ such that
$a+b = 0$

Hence, $(\mathbb{N}, +)$ is not a group.

Ques 2 let $a, b, c \in \mathbb{Z}$

→ **Closure property** :- let $a+b+1 \in \mathbb{Z}$

$\Rightarrow a * b \in \mathbb{Z}$

hence Closed.

→ **Associative property**

$[(a*b)*c] = (a+b+1)*c = a+b+1+c+1 = a+b+c+2$

& $[a*(b*c)] = a*(b+c+1) = a+b+c+1+1 = a+b+c+2$

Hence $[(a*b)*c] = [a*(b*c)]$

$\Rightarrow$ Associative

→ **Identity** :- let $e \in \mathbb{Z}$

$a*e = a \quad \forall a \in \mathbb{Z}$

$a+e+1 = a$ (by defined

$\Rightarrow \boxed{e=-1}$

Hence identity exists

→ **Inverse** :- If $a \in \mathbb{Z}$ then $b \in \mathbb{Z}$ is an inverse of a,

$a*b = -1$

$b+a-1 = -1 \Rightarrow b = -2-a$

Hence inverse exists

→ **Commutativity** :- $a*b = a+b+1 = b+a+1 = b*a$

Hence Commutative

Hence $(\mathbb{Z}, *)$ is an abelian group under $a*b = a+b+1$

Ques 3 Let G be a set of +ve rational numbers. Let $a, b, c \in G$

→ **Closure property** :- let $a, b \in G$

$\Rightarrow a*b = \frac{ab}{2} \in G$: Hence Closed

→ **Associativity** :- $a*(b*c) = a*\left(\frac{bc}{2}\right) = \frac{abc}{4}$

& $(a*b)*c = \left(\frac{ab}{2}\right)*c = \frac{abc}{4}$

Hence, $a*(b*c) = (a*b)*c$

Thus associative

→ **Identity** let $a \in G$ & e be identity element

$$a * e = e * a = a$$

$$\frac{ae}{2} = \frac{ea}{2} = a$$

$\Rightarrow \boxed{e = 2} \in Q^+$ Thus 2 is identity element. — (1)

→ **Inverse** let $a \in G$ then $a^{-1} \in G$

$$a * a^{-1} = e = a^{-1} \times a$$

$$\frac{a a^{-1}}{2} = e = 2 \quad \text{(using (1))}$$

$$a^{-1} = \frac{4}{a} \in Q^*$$

Hence Q^+ is an abelian group under the composition

$$a^* b = \frac{ab}{2}$$

Ques 4

→ **Closure** :- For all $a, b \in S$

$$a * b = (a+b) \bmod 4 \in \{0, 1, 2, 3\} = S$$

Hence Closed

→ **Associativity** :- Since addition modulo 4 is associative

$$(a * b) * c = ((a+b) \bmod 4 + c) \bmod 4 = (a+b+c) \bmod 4$$

$$a * (b * c) = (a + (b+c) \bmod 4) \bmod 4 = (a+b+c) \bmod 4$$

Hence associative

Therefore, $(S, *)$ is Semigroup

→ **Identity** :- let $e \in S$ s.t.

$$a * e = a \quad \forall a \in S$$

$$(a+e) \bmod 4 = a$$

$$\Rightarrow e = 0$$

Also, $a * 0 = (a+0) \bmod 4 = a$

Hence, identity exists

Therefore $(S, *)$ is monoid.

Ques 5 Let us assume, there are two identity element in group G which are e & e'. Then $\forall a \in G$ $e*a = a*e = a$ & $e'*a = a*e' = a$

Since e is an identity element

$\Rightarrow e*e' = e'*e = e'$ $[a*e = e*a = a]$

Now, e' is an identity element

$\Rightarrow e*e' = e'*e = e$ $[a*e' = e'*a = a]$

Thus $e = e*e' = e'*e = e'$

$\Rightarrow e = e'$

Hence unique identity

Ques 6 Let us assume, there are two inverses b & c of element $a \in G$

Then $a*b = b*a = e$ — ① & $a*c = c*a = e$ — ②

where e is an identity element

Now,

$b = b*e$

$b = b*(a*c)$ (substitute $e = a*c$)

$b = (b*a)*c$ (using associativity)

$b = e*c$ (using ① ; $b*a = e$)

$b = c$ (since e is a identity)

Hence, the inverse of an element in a group is unique.

Ques 7 $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

Generators:- An element $a \in \mathbb{Z}_6$ is a generator if $\gcd(a, 6) = 1$

So, $\gcd(1, 6) = 1$

& $\gcd(5, 6) = 1$

Therefore generators are 1, 5

Now,

$1^1 = 1 \bmod 6 = 1$

$1^2 = (1+1) \bmod 6 = 2$

$1^3 = (1+1+1) \bmod 6 = 3$

$1^4 = (1+1+1+1) \bmod 6 = 4$

$15 = (1+1+1+1+1) \bmod 6 = 5$

$\Rightarrow \langle 1 \rangle$ is generator

Similarly $\langle 5 \rangle$ is a generator

Therefore $(\mathbb{Z}_6, +)$ is a cyclic group

Ques let $G = \langle a \rangle$ be a cyclic group generated by a

let $x, y \in G$

then $\exists$ integers m, n, s.t.

$$x = a^m, \quad y = a^n$$

Now

$$xy = a^m a^n = a^{m+n}$$
$$= a^{n+m}$$
$$= a^n \cdot a^m$$
$$= yx$$

$$\Rightarrow xy = yx$$

hence commutative

Therefore G is a abelian group.